Year 12 Maths - Trigonometric functions and equations

Similar questions - Solving equations that become quadratics

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Consider the equation $4\sin x\tan x = 2\cos x + 5$.

(a) Show that the equation can be written as $6\cos^2 x + 5\cos x - 4 = 0$. **[3]**

(b) Hence solve the equation for $0^\circ \leqslant x \leqslant 360^\circ$, explaining why one root of the quadratic gives no solutions. **[4]**

(c) Hence solve $4\sin 2\theta \tan 2\theta = 2\cos 2\theta + 5$ for $0^\circ \leqslant \theta \leqslant 360^\circ$. **[3]**

2 A student is asked to solve $2\cos^2 x + 3\sin x = 3$ for $0^\circ \leqslant x \leqslant 360^\circ$. She writes down $x = 30^\circ$ and $x = 150^\circ$ and says there are no other solutions.

(a) Show that the equation can be written as $2\sin^2 x - 3\sin x + 1 = 0$. **[2]**

(b) Solve the equation and hence explain what the student has missed. **[4]**

Question 1

(a) $\sin x \tan x = \dfrac{\sin^2 x}{\cos x}$, so $4\sin^2 x = 2\cos^2 x + 5\cos x$ after multiplying by $\cos x$. Using $\sin^2 x = 1 - \cos^2 x$: $4 - 4\cos^2 x = 2\cos^2 x + 5\cos x$, so $6\cos^2 x + 5\cos x - 4 = 0$

(b) $(2\cos x - 1)(3\cos x + 4) = 0$. $\cos x = \frac{1}{2}$ or $\cos x = -\frac{4}{3}$; the latter is impossible since $|\cos x| \leqslant 1$. $x = 60^\circ,\ 300^\circ$

(c) $\cos 2\theta = \frac{1}{2}$ with $0^\circ \leqslant 2\theta \leqslant 720^\circ$: $2\theta = 60^\circ, 300^\circ, 420^\circ, 660^\circ$, so $\theta = 30^\circ,\ 150^\circ,\ 210^\circ,\ 330^\circ$

Question 2

(a) $2(1 - \sin^2 x) + 3\sin x = 3$, so $-2\sin^2 x + 3\sin x - 1 = 0$, i.e. $2\sin^2 x - 3\sin x + 1 = 0$

(b) $(2\sin x - 1)(\sin x - 1) = 0$: $\sin x = \frac{1}{2}$ or $\sin x = 1$. $x = 30^\circ, 150^\circ$ and $x = 90^\circ$. She found only the solutions from $\sin x = \frac{1}{2}$ and missed the root $\sin x = 1$, which gives $x = 90^\circ$